

Osama Hamad

Software Engineer — Mobile, Backend & Microservices

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PROFILE

Software engineer who builds production software end to end: Flutter apps shipped to the App Store and Google Play, Laravel and Next.js backends, microservices, and Python automation pipelines. Most of my work is Arabic/RTL-first, built to work offline, and backed by thousands of automated tests and CI gates. Currently pursuing a Master's degree in Artificial Intelligence.

EDUCATION

M.Sc. Artificial Intelligence	In progress
B.Sc. Software Engineering — Islamic University of Gaza	2025

SKILLS

Mobile	Flutter, Dart, Kotlin · GetX, Bloc, Riverpod · Clean Architecture · offline-first sync · Firebase · MyFatoorah, Apple Pay, Google Pay · Google Maps, Mapbox
Backend & Web	Laravel 10/11, Livewire, PHP · Next.js 16, React 19, TypeScript · REST APIs · PostgreSQL, MySQL, Redis, Supabase · Drizzle ORM
Microservices	Independent services communicating over REST · Celery/Redis workers on isolated queues · Docker
AI & Automation	Python, FastAPI, Celery · Google Gemini (text & vision) · OpenCV image processing · Google Sheets / Drive integrations
Security	Certificate pinning, Firebase App Check, MFA, RBAC, Postgres Row Level Security, end-to-end encryption (Argon2id, XChaCha20-Poly1305, X25519)
Quality & Delivery	Unit, widget, golden, integration & e2e testing (Flutter test, PHPUnit, Vitest, Playwright) · GitHub Actions, Codemagic, Fastlane, Docker
Localization	Arabic & English products, RTL-first UX, full i18n
Languages	Arabic (native) · English (good)

SELECTED PROJECTS

Boulevard Super App · Flutter · iOS & Android	2024 – Present
Production multi-vertical super app for the UAE, live on the App Store and Google Play.	
- About 13 verticals in one app (grocery, restaurants, retail, beauty, medical, hotel & flight booking, motors, real estate, insurance and more), with a shared cart, memberships and MyFatoorah payments. 1,500+ Dart files, about 4,900 automated tests and 70+ custom CI guard scripts. Releases go to both stores through Codemagic and Fastlane. - Built the offline cart mutation queue, which survives the app being killed, plus certificate pinning, Firebase App Check, off-main-thread JSON parsing, and p95 startup and frame budgets.	
Stack: Flutter, GetX, Bloc, Firebase, MyFatoorah, Google Maps, UAE Pass, Mixpanel, Codemagic	
Real-Estate Marketplace (UAE) · Flutter	2026
Arabic-first property marketplace: map search, listings, off-plan reservations, short stays, price index, agency subscriptions and an AI assistant.	
- Clean Architecture, with the layering rules enforced as architecture tests in CI. - About 1,800 tests, including golden, RTL and accessibility tests, behind a 70% coverage gate. A Widgetbook design system with 15 catalogs. - In-app card payments with Apple Pay and Google Pay, Mapbox clustering and Firebase phone authentication.	
Stack: Flutter, Bloc, go_router, GetIt/Injectable, freezed, Dio, Mapbox, Widgetbook	

Charity Operations Platform (UK) · Laravel

2025 – 2026

Bilingual Arabic/English platform that runs a charity's operations: case management, events with QR check-in, a school module, volunteers and a public form builder.

- 130 models, 180+ controllers and 258 migrations. Layered into actions, DTOs, repositories and services, with CQRS-style domain modules.
- 320+ PHPUnit test files. CI runs critical-path gates on every change and the full suite nightly.
- RBAC, step-up MFA (TOTP/SMS), CSP headers, rate limiting and encrypted file storage.

Stack: Laravel 10, Livewire 3, MySQL, Redis, Tailwind CSS, Spatie Permission

Argon Lock · Flutter + TypeScript

2026

Zero-knowledge password manager with a Chrome/Edge browser extension (pre-release). Designed the architecture and the security model.

- Argon2id key derivation, XChaCha20-Poly1305 encryption, X25519 device linking, and end-to-end encrypted sync on Supabase.
- Passkeys, TOTP, Android autofill, breach checks using k-anonymity, and duress/decoy vaults.
- The Dart app and the TypeScript extension are checked against the same crypto test vectors, with about 950 tests including tamper and timing tests.

Stack: Flutter, Drift (SQLite), Kotlin, Supabase, TypeScript, Manifest V3, Playwright

Benaa — Offline-First Field App · Flutter + PHP

2025 – 2026

Field app for a humanitarian organisation that registers families and logs home visits, sponsorships and reports without a connection.

- Offline capture into a local Drift/SQLite store with versioned migrations; changed rows are tracked and pushed in paged two-way sync.
- Built both sides: the Flutter app and the PHP/MySQL API (hashed bearer tokens, login lockout, transactional delta-sync endpoint with tombstones).
- Offline sign-in (PBKDF2), Arabic name search with spelling normalisation, PDF/Excel reports. 480+ tests; CI to Firebase App Distribution.

Stack: Flutter, Riverpod, Drift, GoRouter, PHP, MySQL, GitHub Actions

Education Center · Next.js

2026

Management system for a multi-branch tutoring centre (Arabic, RTL, mobile-first): classes, timetables, attendance, fees, payroll, and teacher and parent portals.

- Modular monolith of 16 modules with boundaries enforced by ESLint. Every mutation passes through one pipeline: auth → permission → validation → tenant → audit.
- Each branch's data is isolated in the app layer and by 77 Postgres Row Level Security policies. Database constraints prevent timetable clashes.
- Unit, integration (real Postgres) and Playwright e2e tests. Deployed with Docker and Caddy, with encrypted backups.

Stack: Next.js 16, React 19, TypeScript, PostgreSQL, Drizzle, Better Auth, Tailwind CSS, Playwright

Product Image Automation Pipeline · Python microservices

2026

Automates product images for a grocery e-commerce catalogue: image search, AI verification with Gemini Vision, background removal, normalisation, de-duplication and cloud upload, with a Laravel dashboard where staff curate the results.

- Microservices architecture: the Laravel dashboard calls independent FastAPI services over REST, and Celery workers run on separate queues (crawling, GPU enhancement, embeddings, de-duplication) so each one scales on its own.
- Near-duplicate detection with perceptual hashing and a BK-tree, hybrid search with Reciprocal Rank Fusion, and an image proxy that blocks internal-network requests (SSRF).

Stack: Python, FastAPI, Celery, Redis, Google Gemini, OpenCV, Cloudinary, Laravel 11